

Winning baseball games by solving statistics puzzles

Scott Powers

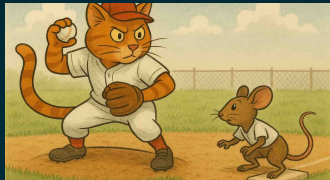
NetSI Talk
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RICE UNIVERSITY
Sport Analytics

Three Puzzles

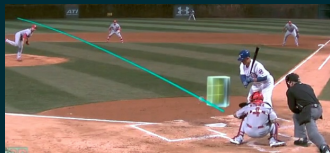
I: How far should you stray from first base?



II: Should you just swing as hard as you can?



III: What does a pitch tell you about a pitcher?



How far should you stray from first base?



Sivaramakrishnan Ramani, Rice University



Jacob Hahn, Rice University → Chicago Cubs



Andrew Schaefer, Rice University

Powers S, Ramani S, Hahn J, Schaefer AJ (2026)

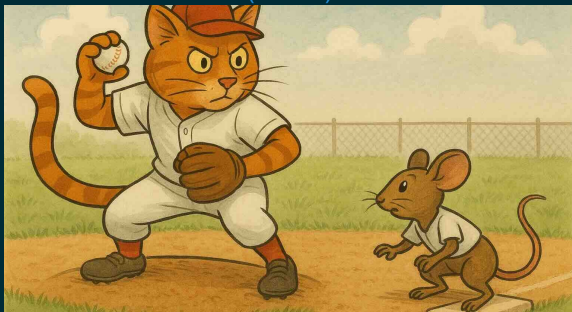
“Lead distance under a pickoff limit in Major League Baseball: A sequential game model”

arXiv:2601.15608

Background

In 2023, MLB limited the pitcher to **3 pickoff attempts**

(video)



After **3 failed pickoffs**, the runner advances automatically

The Puzzle

How should the runner choose their lead distance to maximize **expected runs** scored in the inning?

We got 2022–2023 **pitch-by-pitch lead distance** from MLB*

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Step 1: Estimate the effect of lead distance

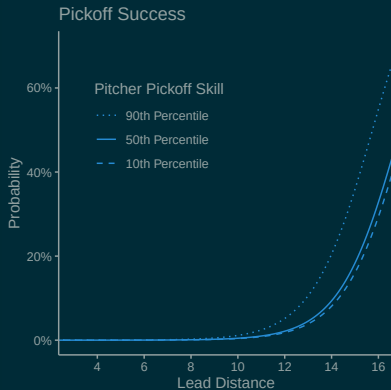
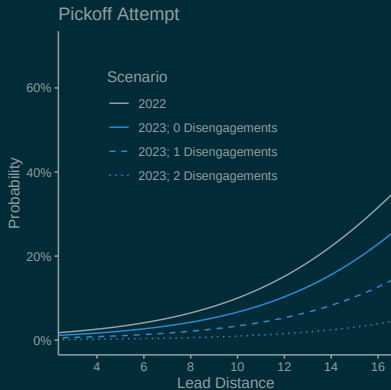
How does lead distance influence...

- **Pickoff attempt** probability?
- **Pickoff success** probability?
- **Stolen base success** probability?

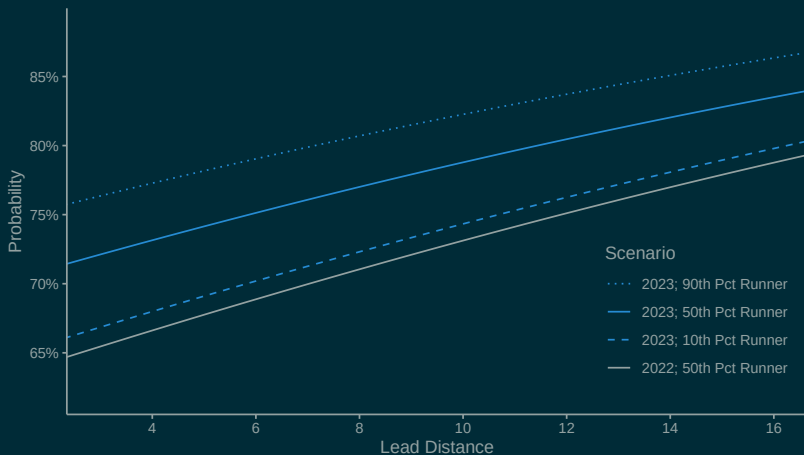
generalized linear mixed-effects model (GLMM)

- binomial likelihood for each outcome
- fixed effects: lead distance, disengagements, out, count, runner sprint speed, catcher arm strength
- random effects: pitcher, runner, catcher

Pickoff attempts increase with lead distance



Stolen bases increase with lead distance



Step 2: Solve for the optimal lead distance policy

What is the **game-theoretic equilibrium** strategy?

two-player sequential game

- state space: bases occupied, outs, count, disengagements
- action space: lead distance (runner), pickoff (pitcher)
- reward function: runs scored
- solved via **value iteration**

The Two-Foot Rule

Count	Disengagements		
	0	1	2
0-0	10.0	11.5	14.1
1-0	10.2	11.7	14.2
2-0	10.2	11.7	14.2
3-0	10.3	11.8	14.2
0-1	9.8	11.4	14.1
1-1	10.0	11.6	14.2
2-1	10.3	11.8	14.3
3-1	10.4	11.9	14.3
0-2	9.5	11.1	14.1
1-2	9.9	11.4	14.3
2-2	10.0	11.5	14.2
3-2	10.4	11.9	14.4

- Increase lead by **two feet** after each disengagement
- Value of implementation:
~12 runs / team / season
(~\$12,000,000)

Should you just swing as hard as you can?



Ron Yurko, Carnegie Mellon University

Powers S, Yurko R (2026) “Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics”
arXiv:2507.01238, to appear in *The American Statistician*

Background

Fans lament batters not having **two-strike approaches**

(video)



HOW LONG IS YOUR SWING?

A SWING'S LENGTH IS CAPTURED FROM THE START OF A SWING UNTIL IMPACT POINT.

A SWING CAN BE AS SHORT AS 4 FEET OR AS LONG AS NEARLY 10 FEET.

The infographic features a white line-art illustration of a batter in mid-swing on a dark blue background. A white curved arrow traces the path of the bat from the start of the swing to the impact point with the ball. The entire graphic is set against a dark blue background with a light blue grid pattern.

STATCAST POWERED BY Google Cloud

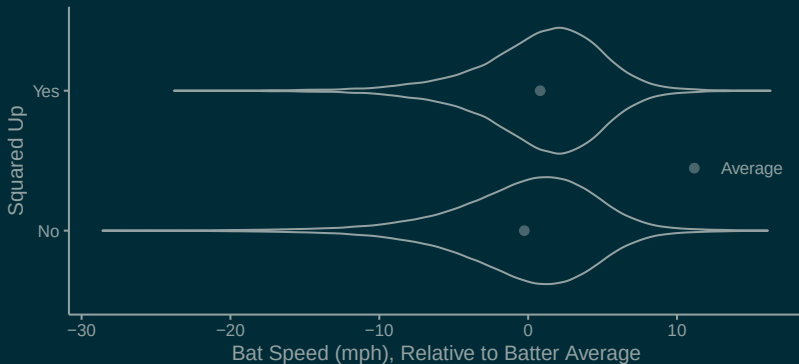
MLB logo

savant

May 2024: MLB releases **swing length** and **bat speed** data

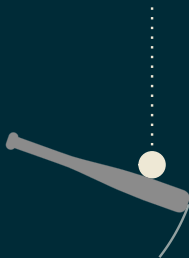
Faster swings make better contact?

"Squared Up" = good contact (exit velocity > 80% of theoretical max)

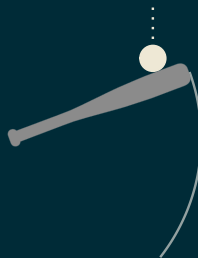


Key Insight

We can't take swing metrics at **face value**



LATE Swing



EARLY Swing

Baseball Savant reports swing metrics at **point of contact**

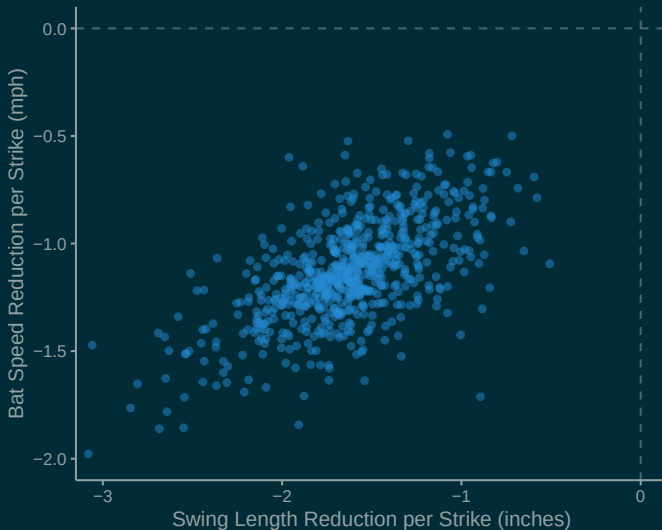
Step 1: Infer batter intention

What does the batter's swing look like
when they make solid contact?

Bayesian hierarchical model

- skew-normal likelihood for bat speed and swing length
- fixed effects: count, location
- random effects: batter, batter $\times$ count, batter $\times$ location

Some batters shorten/slow their swings more



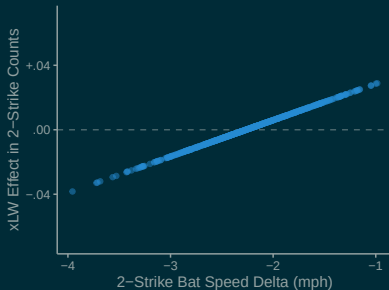
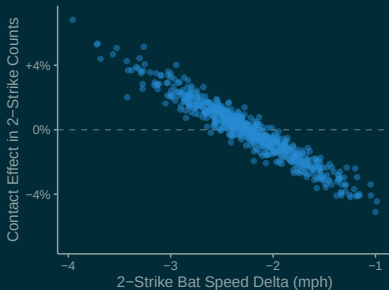
Step 2: Estimate effect of intention

If you make your swing **shorter/slower** with two strikes, do you make **more contact** with two strikes?

Instrumental variables regression (causal inference)

- logistic regression for contact, linear regression for power
- instrument: count
- treatment: batter $\times$ count random effect
- control for difficulty of pitch (gradient boosting model)

Slower swings mean more contact, less power



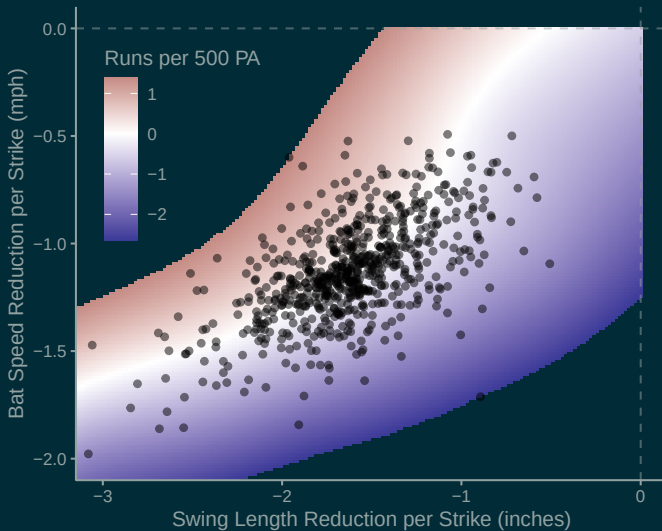
Step 3: Value the contact-power tradeoff

Is **increasing** your contact rate worth **decreasing** your power?

Markov chain model

- state space: ball-strike count
- transition probabilities: depend on approach
- value function: linear weight of plate appearance outcome

The tradeoff is (roughly) balanced



What does a pitch tell you about a pitcher?



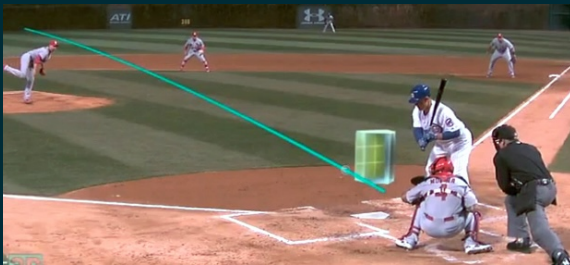
Vicente Iglesias, Susquehanna International Group

Working paper:

Powers S, Iglesias V “Pitch trajectory density estimation for predicting future outcomes”

Background

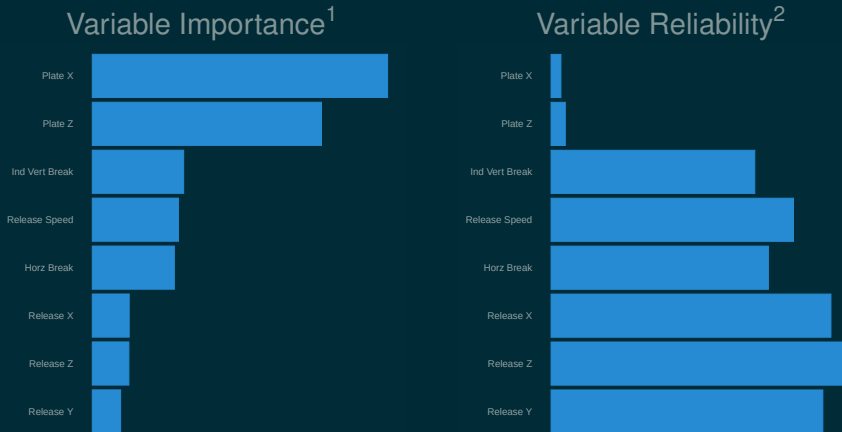
Full **pitch tracking** data have been available for 20 years



Teams evaluate pitchers based on **pitches**, not outcomes

(video #1) (video #2)

Evaluating the *pitch* $\neq$ evaluating the *pitcher*



¹ fractional contribution of each feature's splits to gradient boosting pitch model

² (between-pitcher variance) / (total variance); varies by pitch type (here: RHB FB)

Key Insight

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
...		
x_n	$\rightarrow$	y_n

$$x^* \rightarrow \hat{y} = \hat{f}(x)$$

Our Problem

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
	...		
C	x_n	$\rightarrow$	y_n



$$A \rightarrow \hat{y}$$

Key Insight

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
...		
x_n	$\rightarrow$	y_n

Our Approach

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
	...		
C	x_n	$\rightarrow$	y_n



$$x^* \rightarrow \hat{y} = \hat{f}(x)$$

$$A \quad \hat{p}(x|A) \rightarrow \int \hat{p}(x) \hat{f}(x)$$

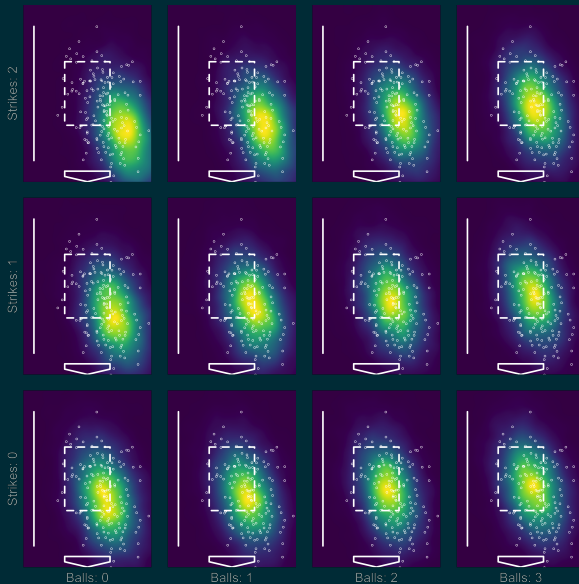
Step 1: Estimate distributions over pitch trajectories

What is the **generative model** for pitch trajectories?

Bayesian hierarchical model

- multivariate normal likelihood (9 dimensions)
- mean $\sim$ pitcher, context, pitcher $\times$ context
- variance scaling $\sim$ pitcher, context
- correlation $\sim$ pitcher

Dylan Cease's Slider vs RHB in All Counts



Step 2: Predict outcomes of pitch trajectories

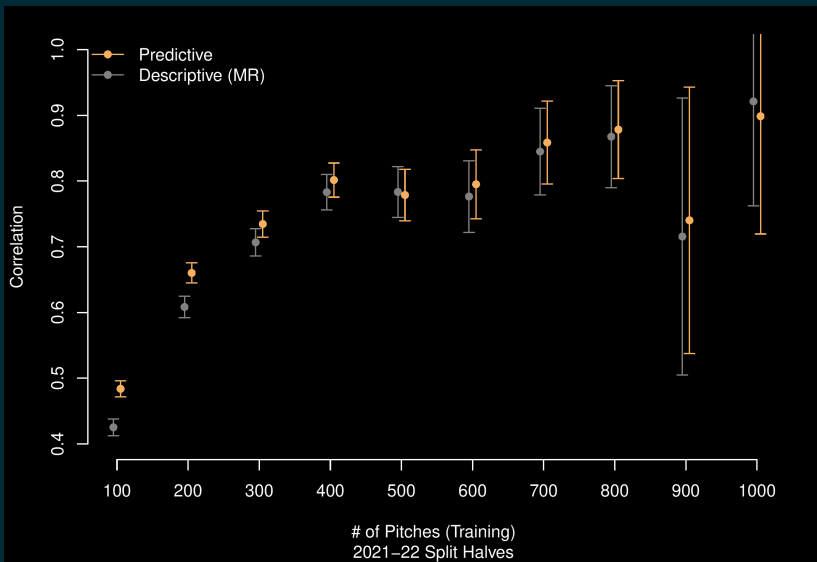
What is the **run value** $\hat{f}(x)$ of each possible trajectory?

gradient boosting

- predict run value of pitch

$$\text{predictive pitch score} = \int \left\{ \hat{p}(x|\text{pitcher, context}) \cdot \hat{f}(x) \right\} dx$$

Predictive pitch score wins for < 300 pitches



Thank You!

saberpowers.github.io/research

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